UNIVERSITY OF CADI AYYAD
FACULTY OF SCIENCES OF TECHNIQUES OF MARRAKECH

Master IAII

Automated Timetable Generation Using Metaheuristic Optimization

Implemented by :
LAALIJI ZAKARIAE
OUADOUD HAMZA
HNIOUA ABDESSAMAD

Supervised by :
Mr. OUAARAB AZIZ

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1 Introduction

The management of pedagogical spaces at the Faculty of Sciences and Techniques of Marrakech (FSTM) is a critical administrative task. It ensures the smooth delivery of various training cycles, including Common Core (TC), Engineering (CI), and Master's programs [1]. Currently, room occupation is managed manually, which is increasingly difficult due to growing student numbers and complex resource interdependencies.

This project aims to design and implement an automated generation system using the **Simulated Annealing** metaheuristic. This approach allows for the navigation of a vast search space to find an assignment of courses, teachers, and rooms that respects all institutional constraints while optimizing for comfort and efficiency [2].

2 Detailed Problem Modeling

To solve the timetabling problem, we must translate real-world requirements into a mathematical optimization framework. We define this as a **Discrete Minimization Problem**.

2.1 Decision Variables and Search Space

The problem is classified as **discrete** because the decision variables—rooms, timeslots, and groups—belong to finite, countable sets rather than continuous ranges. Let C be the set of course sessions to be scheduled. For each session $i \in C$, the algorithm must determine a state vector ω_i :

$$\omega_i = (t, r, p, g)$$

Where :

- $t \in T$: A discrete time slot (e.g., Monday 08 :30–10 :25).
- $r \in R$: A room, categorized by FSTM data as *Amphitheater* (Cap : 240–400), *Large Room* (Cap : 90), or *Small Room* (Cap : 48).
- $p \in P$: A professor from the faculty database.
- $g \in G$: A student group or cohort (e.g., TC S1, MST RD).

2.2 The Objective (Fitness) Function : A Minimization Approach

The core of the modeling lies in the fitness function $f(S)$, which measures the "cost" or "energy" of a given timetable S. Our approach is one of **minimization** ; we seek to reduce the total penalty of constraint violations toward zero.

$$\min f(S) = (W_H \times \sum_{i=1}^{n} H_i) + (W_S \times \sum_{j=1}^{m} S_j)$$

2.2.1 Penalty Weights (W_H, W_S)

To handle the different levels of importance between rules, we use weighting coefficients :

- W_H **(Hard Constraint Weight) :** Assigned a high value (e.g., 1000). This ensures that any physical conflict (like two classes in one room) creates a massive penalty, forcing the algorithm to prioritize feasibility.
- W_S **(Soft Constraint Weight) :** Assigned a low value (e.g., 1 or 10). These weights represent "preferences" (like avoiding gaps in a student's day) to differentiate between feasible schedules.

2.3 Constraint Formalization

The variables H_i and S_j represent the counts of violations :
Hard Constraints (H) :

- *Overlap Constraint :* $\sum Assignments(p,t) \leq 1$ and $\sum Assignments(r,t) \leq 1$.
- *Capacity Match :* $\forall \omega_i, Capacity(r) \geq Size(g)$.
- *Resource Specificity :* Modules requiring Amphitheaters cannot be placed in small rooms.

Soft Constraints (S) :

- *Contiguity :* Minimizing the idle time ("holes") between two sessions for the same group g.
- *Distribution :* Spreading the pedagogical load evenly across the working week.

3 Methodology : Simulated Annealing

Simulated Annealing (SA) is chosen for its ability to navigate discrete search spaces and its efficiency in escaping local optima through probabilistic state transitions.

3.1 Algorithmic Process

1. **Initialization :** Generate an initial random assignment of all courses.
2. **Neighbor Generation :** Perform a "move" by swapping the timeslots of two courses or reassigning a course to a different room.
3. **Acceptance Criteria :**
   - If the new state S' has a lower cost ($\Delta f < 0$), accept it.
   - If S' is worse ($\Delta f > 0$), accept it with probability $P = e^{-\frac{\Delta f}{T}}$.
4. **Cooling Schedule :** Gradually reduce T following $T_{new} = \alpha T_{old}$ (where $\alpha \approx 0.95$). As T approaches zero, the algorithm "freezes" into a global minimum.

Références

[1] Faculty of Sciences and Techniques of Marrakech (FSTM). *Occupation des locaux - Session Automne 2025-2026*, 2025.

[2] Aziz Ouaarab. *Project Statement : Automated Timetable Generation Using Metaheuristic Optimization*. FSTM, 2025.